

PAPER_1373 — Vacuum Birefringence (Schwinger Pair)

Framework: UQFF — Star-Magic v5.27+ | **Tier:** H Applied/Engineering (CLOSED) **Calculator:** `calculate_paradox({"paradox": "vacuum_birefringence"})`

Master Expression

Threshold = $\Lambda^2 \times E_{\text{Schwinger}} \approx 7 \times 10^{13} \text{ V/m}$

Match: below Schwinger 1.3×10^{18}

Interpretation

UQFF predicts birefringence onset at Λ^2 suppression below Schwinger. PVLAS-2024 null consistent. Detectable at next-gen high-E lasers.

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$
- $\omega_{\text{SCm}} = 1.25 \text{ THz}$ ($\alpha = \Lambda$ closed ledger identity)

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

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